

Overview

Extending Workday Prism Beyond Native Cloud Storage Limitations – Oracle Cloud Storage Integration Using Workday Studio

In this example, we will walk through the end-to-end setup and configuration required to integrate Oracle Cloud Infrastructure (OCI) Object Storage with Workday Studio. The implementation demonstrated in this document was performed using an OCI Free Tier subscription.

The following sections describe the prerequisites, OCI configurations, and the details required to establish secure communication between Workday and Oracle Cloud Object Storage.

Steps Involved

1. Create an OCI Free Tier Account
2. Gather the Required OCI Details:
 - a. Tenancy OCI ID
 - b. User OCI ID
 - c. Object Storage Namespace
 - d. Private Key
 - e. Key Fingerprint
 - f. Bucket Name
 - g. Object Name

These details will be used within Workday Studio to construct the signed OCI REST API requests, authenticate securely using RSA-based signing, and retrieve file data from Oracle Cloud Object Storage.

OCI - Create Bucket and Upload File

al perform
service f

Profile

deepakku...@gmail.com

Identity domain: Default

Tenancy: deepakku...

Language: English

User settings

Console settings

Sign out

← Tenancy details

deepakkum... Active

Tenancy

Details Work requests Tags

General Information

1 - Tenancy OCI ID

OCID	ocid1.tenancy.oc1..aaaaaaaaxzrticjdwo36smyeudn...	Copy
Name	deepakkum...	
Audit retention period days	365	
Home region	LHR	
CSI number	126283010	
Amazon S3 compatibility API designated compartment	deepakkum...	
SWIFT API designated compartment	deepakkum...	
Object storage namespace	lrrOxeroxrw	3 - Object Storage Namespace

- Q

Search
- Home

Compute

Storage

Networking

Oracle AI Database

Databases

Analytics & AI

Storage

Block Storage

- Block Volumes
- Block Volume Backups
- Block Volume Replicas
- Volume Groups
- Volume Group Backups
- Volume Group Replicas
- Backup Policies

File Storage

- File Systems
- Mount Targets

Lustre File Storage

- Lustre File Systems

Object Storage & Archive Storage

- buckets
- Private Endpoints

Q

Search and Filter

Applied filters

Compartment **deepakkur**

Create bucket

View or manage logs

Create bucket

Bucket name

workday-integration-test

Bucket scope

Namespace

Default storage tier

Standard

Enable auto-tiering

Applied filters

Compartment **deepakkumarsray2026 (root)**

Create bucket

View or manage logs

Name ^	6 - Bucket Name	Default storage tier	Visibility	Time created
	workday-integration-test	Standard	Private	May 6, 2026, 05:04 UTC

Details

Objects

Management

Monitoring

Policies

Work requests

Tags

Objects

Q Search object by name or prefix

Upload objects

Actions ▼

Upload objects

1

Select files

Required

Object name prefix
samplefile

Storage tier
Standard ▼

Additional checksum
None ▼

Choose Files from your Computer

Drop a file or select one

Select or drop files here

File upload

7 - Object Name

Clear

wdsamplefile.csv

40 B

×

Basic info

Object name prefix	samplefile
Storage tier	Standard
Additional checksum	None

Files to be upload

Files to be upload	Size
wdsamplefile.csv	40 bytes

Page 1 of 1 (1 - 1 of 1 total items) Items per page 10

Upload objects



Select files
Required



Review and upload files

Uploading files to object store

64 MiB and smaller file uploads
samplefilewdsamplefile.csv

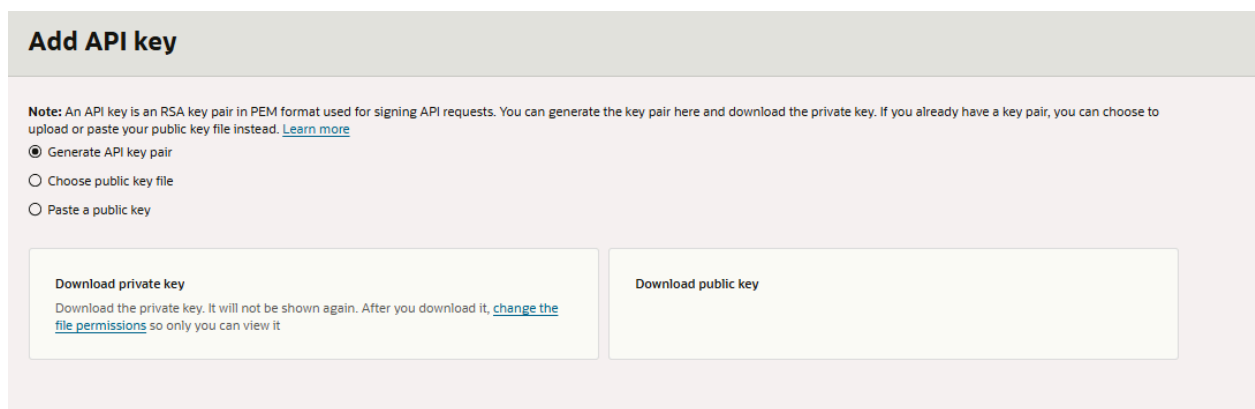
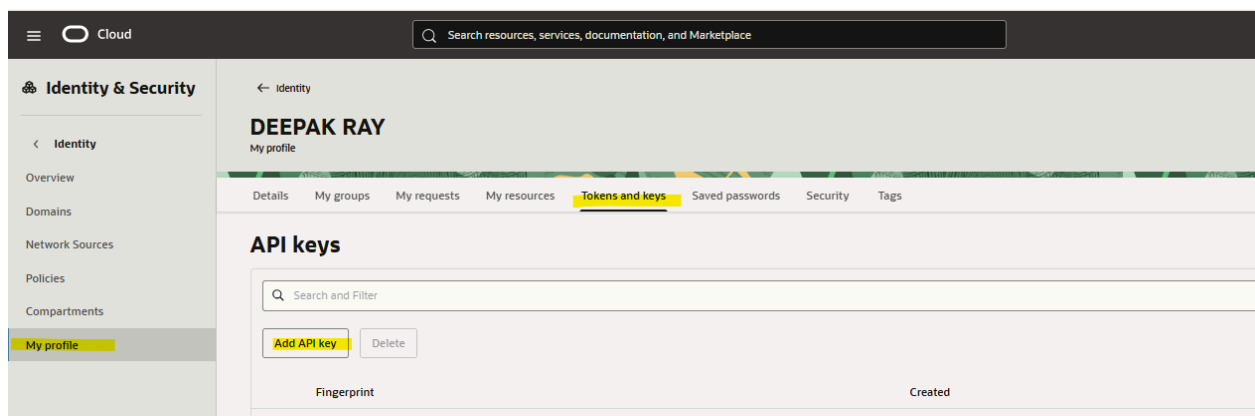
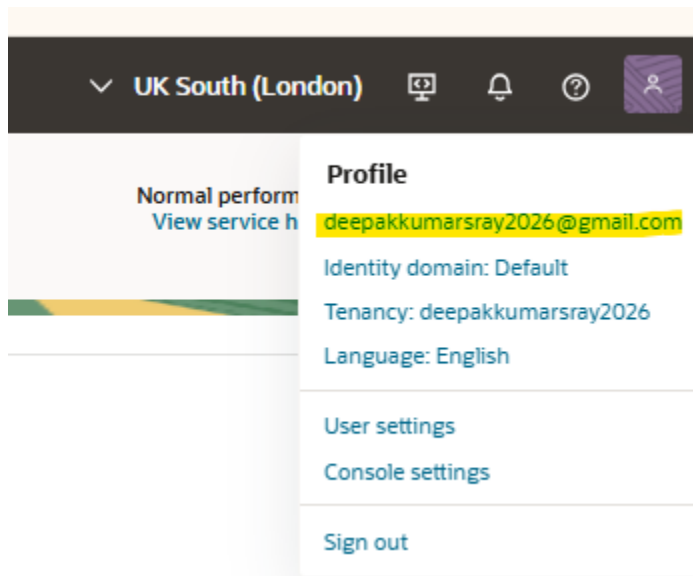
100%  Done

Important - Assign appropriate security policy to the bucket.

OCI - Create API Signing Key

OCI - Create API Signing Key

Create the Private key and Finger print which will be required for integration.



Add API key

Note: An API key is an RSA key pair in PEM format used for signing API requests. You can generate the key pair here and download the private key. If you already have a key pair, you can choose to upload or paste your public key file instead. [Learn more](#)

☒ Generate API key pair

☐ Choose public key file

☐ Paste a public key

Download private key

Download the private key. It will not be shown again. After you download it, [change the file permissions](#) so only you can view it.

Download public key

Close

Add

Download the private key and save it correctly. There may be additional text like - OCI_API_KEY, if present then delete those lines.

Configuration file preview

Note: This configuration file snippet includes the basic authentication information you'll need to use the SDK, CLI, or other OCI developer tool. Paste the contents of the text box into your `~/.oci/config` file and update the `key_file` parameter with the file path to your private key. [Learn more](#)

Fingerprint
2c:70:1e:08:7a:c6:39:2c:70:ae:26:f7:78:ac:5c:f1

```
[DEFAULT]
user=ocid1.user.oc1..aaaaaaaakxahnvdvu2kyqnukodfzyqhmas5if3q4t2ktuxbhdchhhxeosk3a
fingerprint=2c:70:1e:08:7a:c6:39:2c:70:ae:26:f7:78:ac:5c:f1
tenancy=ocid1.tenancy.oc1..aaaaaaaaxtrtrcjgdwa36sryeudn7auu7am1egs15bxs11mqj0jt24vcpa
region=uk-london-1
key_files=<path to your private keyfile> # TODO
```

4 - Private Key

Configuration file preview [DEFAULT] user=ocid1.user.oc1..aaaaaaaakxahnvdvu2kyqnukodfzyqhmas5if3q4t2ktuxbhdchhhxeosk3a fingerprint=2c:70:1e:08:7a:c6:39:2c:70:ae:26:f7:78:ac:5c:f1

Copy

← identity

DEEPAK RAY
My profile

Actions Edit my profile

DetailsMy groupsMy requestsMy resourcesTokens and keysSaved passwordsSecurityTags

API keys

☐	Fingerprint	Created	
☐	2c:70:1e:08:7a:c6:39:2c:70:ae:26:f7:78:ac:5c:f1	May 06, 2026, 05:25:49 UTC	...

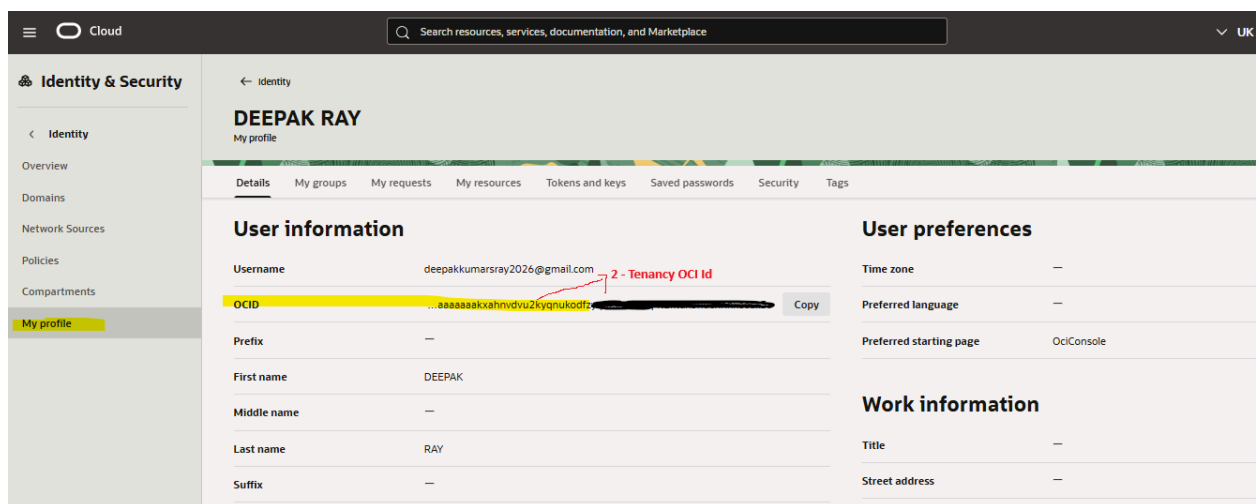
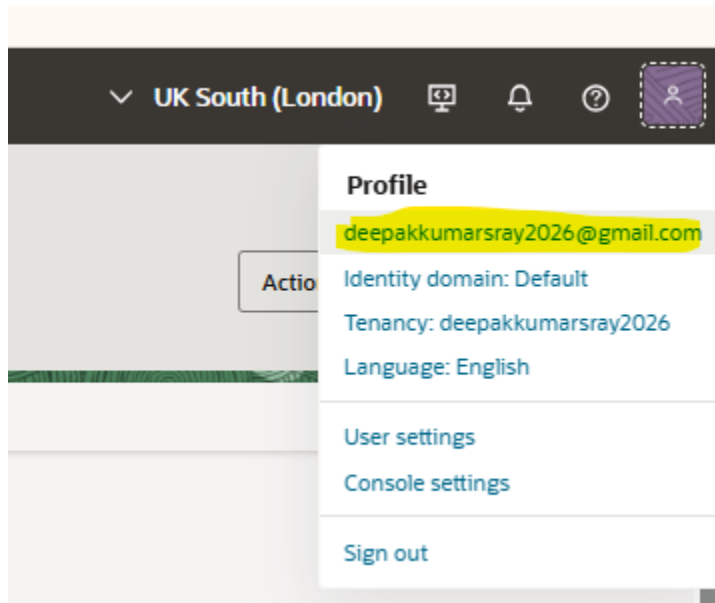
Page 1 of 1 (1 - 1 of 1 total items)

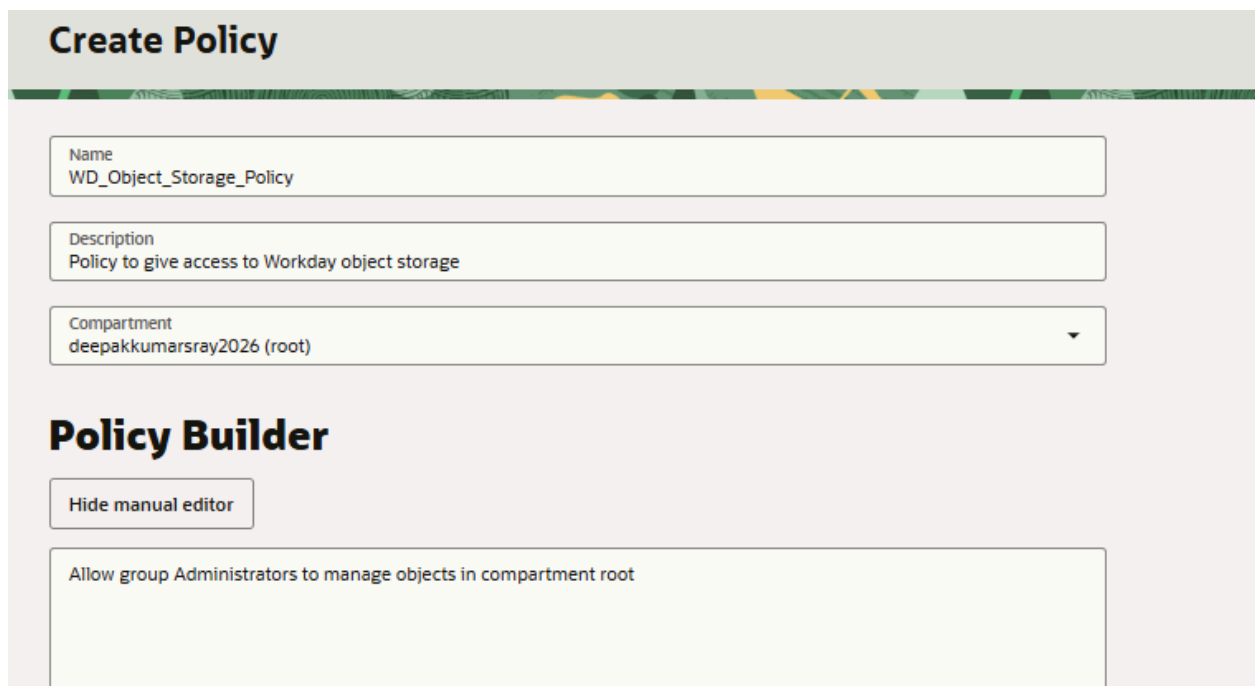
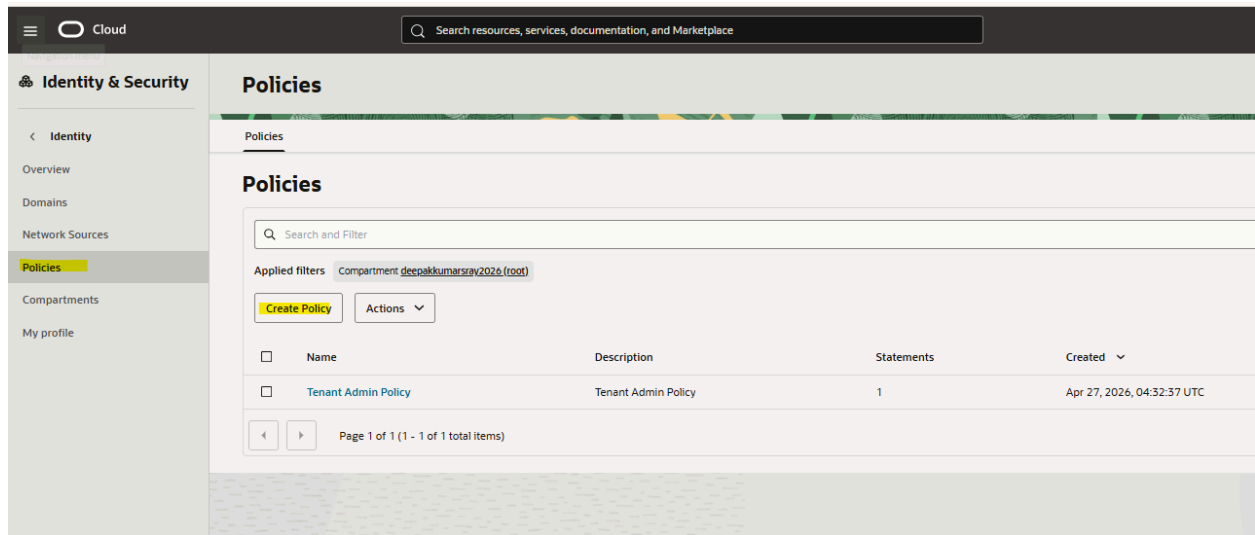
Items per page 10

5 - Fingerprint

OCI - IAM Policy for Object Storage

Click on





This is not the right way to assign the security policy. Just for the POC this level of access is given.

WD Studio Integration

Workday Integration Attributes

Integration Attributes

Security

Integration Attributes 9 items

Attribute Provider	Attribute	Description	Value
Cloud Integration Broker	WorkdayIn Connector ID		DKR_OCL_Connection/StartHere
DKR_OCL_Connection	oci.tenancy.ociid		ocid1.tenancy.oc1..aaaaaaaaaxztrcjjdwo36smyeudn7auujamiagsi5bxsiimqyjojt24vcpa
	oci.user.ociid		ocid1.user.oc1..aaaaaaaaakxahnvdvu2kyqnukodfzyqhmas5lf3q4t2ktuxbhdchhhxeosk3a
	oci.fingerprint		2c:70:1e:08:7a:c6:39:2c:70:ae:26:f7:78:ac:5c:f1

Integration Attributes 9 items

Attribute Provider	Attribute	Description	Value	Restricted to Environment
	oci.private.key		-----BEGIN PRIVATE KEY----- MIIEvQIBADANBgkqhkiG9w0BAQEFAASCBKcwggSjAgEAAoIBAQCprQGjFAojWjob wyxVSXOCcb24b0zyqtsd31vGW2ui4QY9kpiEnOB8Lfp3m4rPq1OIF/nO1bfM9kLI nOQSS7hTfNqzbwRFcOTyNj0+G6bmmnf+bZX2M0Dvx8SLFwFhdoJtg0NBAAw9JS Mr GeRK9w1BZFETlQpcWAjITk/fmVhWFT1Xb8PDNE06sZFDgFD8NJ ovWyG9Abc/iNjag qZ9cuZjT2EFfjW6UEDJleshfCXVqhV50r19SHAz bFv6LtM8x2RtjbnB3Zn1Fd0Jt EfzQaCeYADUHMLL9YsoB4PpMEIZp XJu2Vwd+5hzfeOj6cUPex8TkonLmc8fEtirV JcTUS0gBAGMBAAECg gEAAQ5tB3o2tBxyZr/8P5qD+idM7t5PvsJMBi6g4ISHfoll m+0qH3Tn 9pn0ZsFB6bBSPy/QBqbdR0lInn3AnzeY8ds61yHer2K2NGcnhzAhCra 1i 0jil2mfVXbRaOlRmC3RqxfyvZovLUmD2/dKiQ1hiGF0ibqrhzewt4Zi UxgbrP/8s rgVbiftNfFdOnFdOnFdId8BtE8hRTfIU4K5md8E90tkodXGi b/OzJhjzZigWGTe/ RVmILJluHdNas6SKljin6Dxy9FcK8yfpglbnRxAk T6MYRiv45fiFZiwxfd5VYa EZNY0npYf/5ggMg6tGTSQgKJslZRZZGil pf+yUtrRuQKBgQDXEa+OP0Gz69wRuWjS SYcYy7tkca0PX4S/11ZZ9 B/84Vvgtc7mN7zB/DSpELDAusYJQ+vYzlHNuD9PSg qM3VXFp8o b4LdL/Sw95f2vnJzQrTbdb7nycdgQ7QYdOfAn8SY2ki6quEA4y5o6m s Sx4chQC+Yi8Ps+KvTSZOiA53qQKBgQDj97ptA+UujNOswGvPxML WtLQJgBXPXC4f zeH80SeyW5kt8H58F5xIRKmWFXLSUIPqWVT0nT RS/M8mX7a5WD+w7YQXPlgz7Hzb 2kJ8IDnG98X1ehyTwEo4RsVm 0oaJOso3JmrTAfDwtngNKL6OD9wt3ofeTtY4fdXq 2Nw7X4lkmQKB -----	
	oci.namespace		4KesaUy44GDfdRX4z90aNE8+ 1MuTRiEBxYGHOFB/K06aW1Y	
	oci.bucketname		workday-integration-test	
	oci.objectname		samplefilewdsamplefile.csv	

Integration Attributes 7 items

Attribute Provider	Attribute	Description	Attribute Value(s)	
			Value	Restricted to Environment
Cloud Integration Broker	WorkdayIn Connector ID		DKR_OCI_Connection/StartHere	
DKR_OCI_Connection	oci.tenancy.oid		ocid1.tenancy.oc1..aaaaaaaaxztrcjddwo36smyeud bxslmqjajt24vcpa	
	oci.user.oid		ocid1.user.oc1..aaaaaaaakxahnyd uxbhdchhhxeosk3a	
	oci.fingerprint		2c:70:1e:08: 5c:f1	
	oci.region		uk-london-1	
	oci.private.key		-----BEGIN PRIVATE KEY----- MIIEvQIBADANBgkqhkiG9w0BAQEFAASCBKcwggSjAgEAAoIBAQCprQGjFAoJWjob wyxVSXOCcb24b0zyqtsd31vGW2ui4QY zbuwRFeOTy Mr GeRK9w1BZFeTIQpcWAjITk/fmVhWF1Xb8PDNE06sZFDgFD8NJovWyG9Abc/INJag qZ9cuZJT2EFRjW6UEDJleshfCXVqhV50r19SHAz	

	oci.namespace		lrtr0xeroxrw
--	---------------	--	--------------

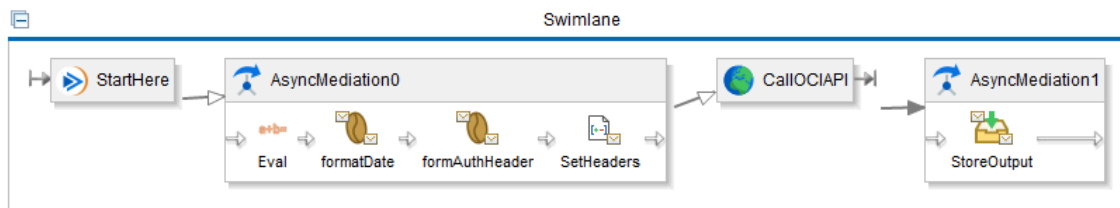
Launch Parameters

Integration Attributes **Launch Parameters** Security

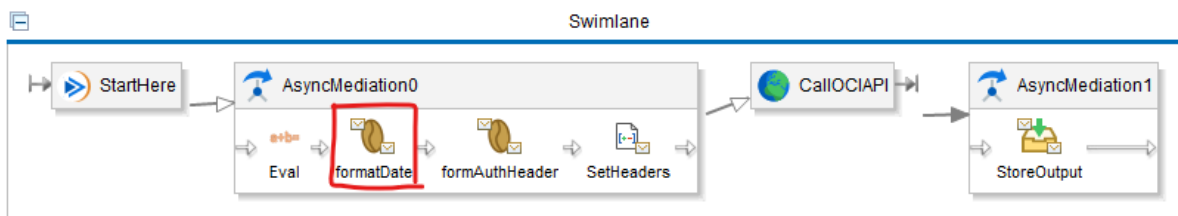
Launch Parameters 2 items

Launch Parameter	Name	Launch Parameter Type
Q	bucketname	Data Type T Text
Q	objectname	Data Type T Text

Workday Studio Assembly



Fetch the integration attributes and launch parameters



Expression

```

props['oci.tenancy.oid']=intsys.getAttribute('oci.tenancy.oid')
props['oci.user.oid']=intsys.getAttribute('oci.user.oid')
props['oci.fingerprint']=intsys.getAttribute('oci.fingerprint')
props['oci.region']=intsys.getAttribute('oci.region')
props['oci.private.key']=intsys.getAttribute('oci.private.key')
props['oci.namespace']=intsys.getAttribute('oci.namespace')
//props['oci.bucketname']=intsys.getAttribute('oci.bucketname')
props['oci.bucketname']=lp.getSimpleData('bucketname')
//props['oci.objectname']=intsys.getAttribute('oci.objectname')
props['oci.objectname']=lp.getSimpleData('objectname')

```

Form the complete OCI endpoint and keyid

```

props['oci.host']="objectstorage." + props['oci.region'] + ".oraclecloud.com"
props['oci.path']="/n/" + props['oci.namespace'] + "/b/" + props['oci.bucketname'] + "/o/" + props['oci.objectname']
props['oci.urlStr'] = "https://" + props['oci.host'] + props['oci.path']
props['oci.keyId'] = props['oci.tenancy.oid'] + "/" + props['oci.user.oid'] + "/" + props['oci.fingerprint']

```

Custom Java program to form the date (this can be done in Studio's MVEL too)

```
package com.oci;

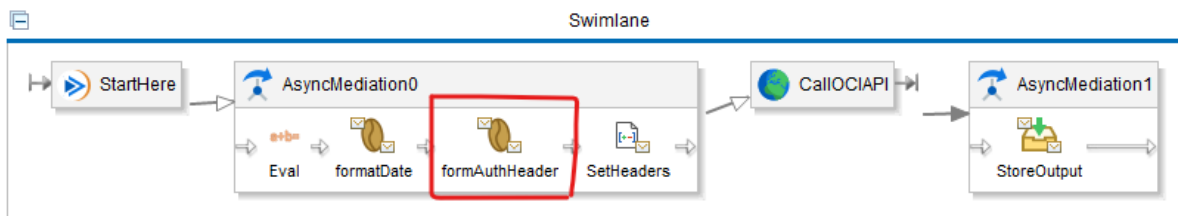
import com.capeclear.assembly.annotation.Component;

/**
 * Custom mediation
 *
 * TODO Modify the Component annotation. Also add Property annotations to any
 * bean pattern methods you add and want to appear in the Assembly Editor.
 */
@Component(name = "formatdate", type = mediation, tooltip = "", scope = "prototype", smallIconPath = "icons/formatdate_1")
public class formatdate {

    @Property(name = "mcFormatDate")
    @ComponentMethod
    public void process(com.capeclear.mediation.MediationContext arg0) {
        // TODO implement
        SimpleDateFormat sdf = new SimpleDateFormat("EEE, dd MMM yyyy HH:mm:ss z", Locale.US);
        sdf.setTimeZone(TimeZone.getTimeZone("GMT"));
        String dateStr = sdf.format(new Date());

        arg0.setProperty("mcFormatDate", dateStr);
    }
}
```

Second custom Java component to form the authorisation string



```

@property(name = "mcAuthHeader")
@ComponentMethod
public void process(com.capeclear.mediation.MediationContext arg0) {
    // TODO implement
    // =====
    // ENTRY POINT
    // Studio calls this block. Input variables are pre-populated from
    // the mediation. Output variables are written back.
    // =====

    String fileContentBase64 = "";
    String contentType = "";
    String httpStatus = "";
    String errorMessage = "";

    String region = (String)arg0.getProperty("oci.region");
    String namespace = (String)arg0.getProperty("oci.namespace");
    String tenancyOcid = (String)arg0.getProperty("oci.tenancy.ocid");
    String userOcid = (String)arg0.getProperty("oci.user.ocid");
    String fingerprint = (String)arg0.getProperty("oci.fingerprint");
    String privateKeyPem = (String)arg0.getProperty("oci.private.key");

    String bucketName=(String)arg0.getProperty("oci.bucketname");
    String objectName=(String)arg0.getProperty("oci.objectname");

    String host = (String)arg0.getProperty("oci.host");
    String path = (String)arg0.getProperty("oci.path");
    String urlStr = (String)arg0.getProperty("oci.urlStr");
    String keyId = (String)arg0.getProperty("oci.keyId");

    String dateStr = (String)arg0.getProperty("mcFormatDate");

```

```

try {
    // -----
    // 1. Load the private key from the PEM string
    //    (passed in from the encrypted Integration Attribute)
    // -----
    PrivateKey privateKey = loadPrivateKey(privateKeyPem);

    // -----
    // 2. Build the signing string
    //    For GET: date, (request-target), host
    // -----
    String requestTarget = "get " + path;
    String signingString = "date: " + dateStr + "\n"
        + "(request-target): " + requestTarget + "\n"
        + "host: " + host;

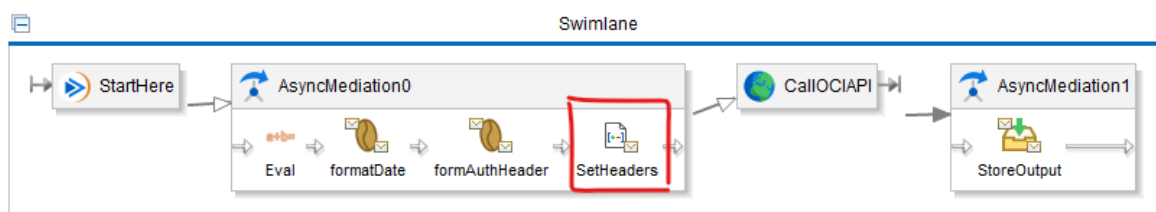
    // -----
    // 3. Compute the RSA-SHA256 signature
    // -----
    Signature sig = Signature.getInstance("SHA256withRSA");
    sig.initSign(privateKey);
    sig.update(signingString.getBytes("UTF-8"));
    byte[] signedBytes = sig.sign();
    String signature = Base64.getEncoder().encodeToString(signedBytes);

    // -----
    // 4. Build the Authorization header
    // -----
    String authHeader = "Signature version=\"1\", "
        + "keyId=\"" + keyId + "\", "
        + "algorithm=\"rsa-sha256\", "
        + "headers=\"date (request-target) host\", "
        + "signature=\"" + signature + "\"";
    System.out.println(authHeader);
    arg0.setProperty("mcAuthHeader", authHeader);
}

```

mcAuthHeader property will save the required authorization string.

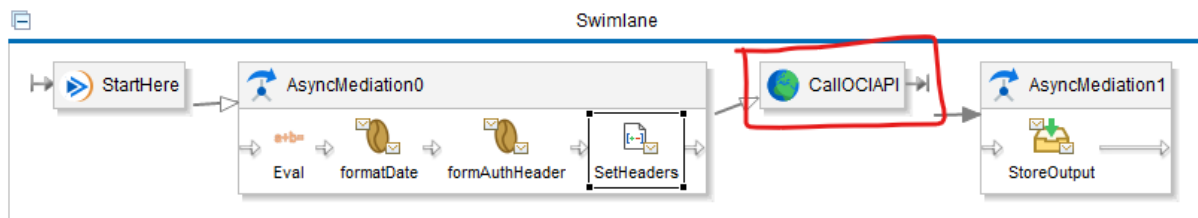
Set API Headers



Add Headers

Name	Value	
date	@{props['mcFormatDate']}	↑ ↓ ✕
host	@{props['oci.host']}	↑ ↓ ✕
Authorization	@{props['mcAuthHeader']}	↑ ↓ ✕

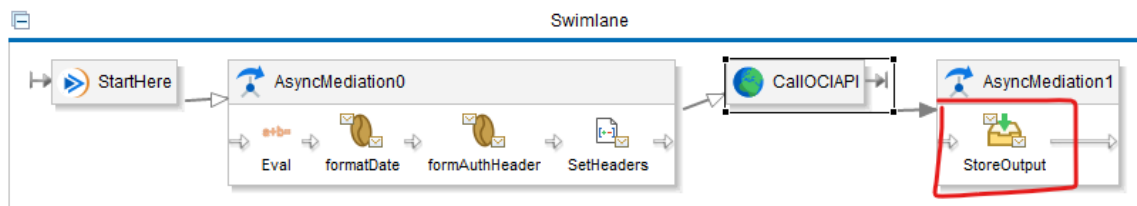
Call the API



http-out

Common	Id	CallIOCI API
Advanced	Routes Response To	AsyncMediation1
	Execute When	Press ctrl-space for content assist
	Endpoint	@{props['oci.urlStr']}
	Http Method	GET

And finally, store the output.



Once the file is retrieved, it can then be passed on to next BP's for processin.